Hydrophoretic high-throughput selection of platelets in physiological shear-stress range†

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A gentle, but fast means for low-stress, high-throughput platelet purification is of significant clinical and biotechnological utility. Current implementations to sort platelets, however, require an external physical field, specialized buffer, or the harsh separation condition of high shear stress that tends to cause platelet stimulation. Here we report the use of hydrophoretic size separation in a wider channel and its parallelization to augment its throughput capability, maintaining physiological shear-stress range. We demonstrate a parallelized device comprising 10 stacks of the wide-channel hydrophoresis device, yielding a throughput of 2.9 million cells s^{-1} and a platelet purity of 76.8%. The use of the wide channel for hydrophoresis also facilitates clogging-free separation by sorting blood clots and plaques. The wide-channel hydrophoresis offers the potential for gentle, fast, clogging-free sorting of rare blood cells with extreme throughput capabilities.

Introduction

Platelets are small, disc-shaped, anuclear blood cell fragments that are responsible for hemostasis by formation of a blood clot and secretion of growth factors. These simplified cells that lack a nucleus are known for thwarting blood loss, but they also play a significant role in many pathophysiological processes such as thrombosis, hemorrhage, inflammation, antimicrobial host defense, tumor growth, and metastasis.[1–3] Thus, platelet alterations are strongly associated with distinct clinical diseases, including cardiovascular diseases, diabetes, tumor metastasis, and sepsis.[4,5] In recent years, additional platelet functions have continued to come to light. Platelets can be drug targets in sepsis, and also foster cancer and rheumatoid arthritis.[2] However, because blood is a very complex mixture containing red blood cells (RBCs), white blood cells (WBCs), and platelets, blood samples need to be separated prior to analysis and further application. In addition, platelets are relatively fragile so that they can be easily stimulated by high shear stress and temperature change during blood collection and sample preparation, thereby causing artifactual platelet activation.[6,7] Therefore, the development of high-purity, gentle, but fast means to sort platelets from other blood components is essential to unveil their pathophysiological functions and facilitate the understanding their relationship with distinct clinical diseases.

Physical separation methods are capable of purifying platelets from a heterogeneous cell suspension of blood by using the difference in cell size, shape, or density. Centrifugation is routinely used for platelet purification that the buffy coat containing most of platelets and WBCs is sandwiched between blood plasma and above RBCs.[8] Although there are reliable protocols to prepare platelet-rich plasma (PRP) by centrifuge, most of them require high-speed centrifugation of above 3000 *g* that can result in significant cell loss.[9,10] Microfluidic separation methods also enable platelet purification by utilizing physical forces or hydrodynamic phenomena. Acoustic forces have been used for continuous separation of platelets and RBCs by adjusting the density of the suspending medium with caesium chloride.[11] Dielectrophoresis (DEP) has been employed for the selection of platelets by exploiting the dependence of the DEP force on a cell volume.[12] It also requires a specialized buffer of low electrical conductivity to maintain high field strength without electrolysis. Although they showed impressive results, they require external physical fields and specialized buffers that can have negative effects on cell viability and even cause platelet activation. Inertial forces have been used for platelet separation by utilizing an intrinsic hydrodynamic phenomenon, the size dependence of inertial ordering.[13] This method enables high-throughput separation, yielding a throughput up to ≈1.5 million cells s^{-1} without external fields and medium exchange. There is still a challenge for the development of a gentle, but fast method for low-stress, high-throughput platelet purification.

To address the above need, we present here a simple and effective method to achieve a purified platelet population in physiological shear-stress range, called wide-channel hydrophoresis. We designed a hydrophoresis channel[14–17] with a wide width in order to maintain physiological shear-stress range even for high-throughput separation of platelets, thereby preventing platelet stimulation. The hydrophoresis channel enables efficient separation by utilizing convective, rotational flows induced from

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regularly patterned anisotropic microfluidic obstacles. Its separation performance is characterized experimentally using 4 μm beads and blood cells. We also demonstrate the robustness of the hydrophoretic method for practical applications by sorting platelets in a parallelized sorting device to augment the throughput capability up to 10-fold.

Experimental

Design and fabrication of wide-channel hydrophoresis device

We fabricated the mold structures with two-step photolithography in SU-8 photoresist (SU8-2010; Microchem Corp., MA): The first layer of photolithography defined the linear-channel structure (8 μm in depth); the second layer was aligned to lie on top of the linear-channel structure in the first layer and defined the pattern of slant grooves (14 μm in depth). To ensure complete cell ordering, we placed 530 grooves, more than the number of the previous devices, at least 120 grooves. The device was microfabricated in poly(dimethylsiloxane) (PDMS) by using soft lithography as shown in Fig. 1, in which the channel width W was 1000 μm, the channel height $H = 22$ μm, the gap height of the obstacles $H_{ob} = 8$ μm, the slanting angle of the obstacles relative to the sidewall 2 and bulk flow $\theta = 80°$, the obstacle thickness $L_{ob} = 27$ μm, and the pitch distance between the obstacles $L = 33$ μm (Fig. 2). The PDMS microchannel was then irreversibly sealed by plasma activation on a glass slide. The deviations of all dimensions were less than 5%. To augment the throughput capability of the wide-channel hydrophoresis, the parallelized sorting device was fabricated by repetitively bonding a channel layer on top of the other layer up to 10 layers.

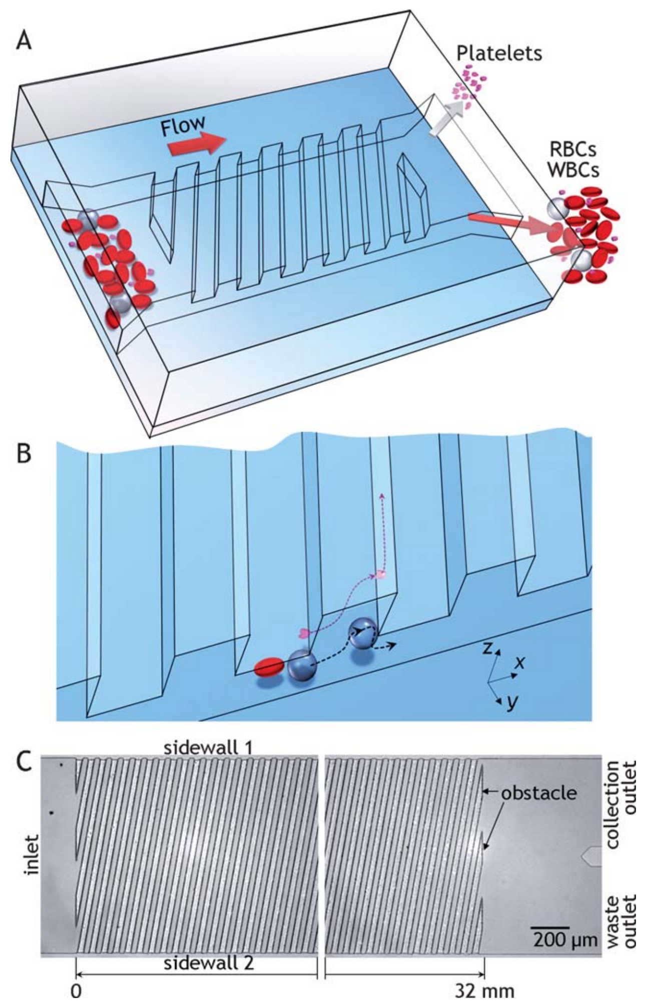

Fig. 1 Size separation of platelets in a wide-channel hydrophoresis device. (A) Schematic illustrating the hydrophoresis device with anisotropic obstacles and its separation of platelets. (B) The larger cells (RBCs and WBCs) than the critical threshold stay near the sidewall 2 by a steric hindrance mechanism and the smaller cells (platelets) remain unfocused, separated from the larger cells. (C) The hydrophoresis device consists of 530 anisotropic obstacles slanted at an angle of 80° relative to the sidewall 2 and bulk flow.

Material preparation

Red fluorescent polystyrene beads with a diameter of 4 μm were purchased from Molecular Probes (Eugene, OR) only to characterize the ordering criteria of the wide-channel hydrophoresis device. The beads were prepared in 2% Pluronic F68 solution (Sigma–Aldrich, St. Louis, MO). For platelet separation, Sprague-Dawley rat blood was drawn with ethylenediaminetetraacetic acid (EDTA) as anticoagulant. Blood samples were used within 8 h after collection. The concentration of blood cells

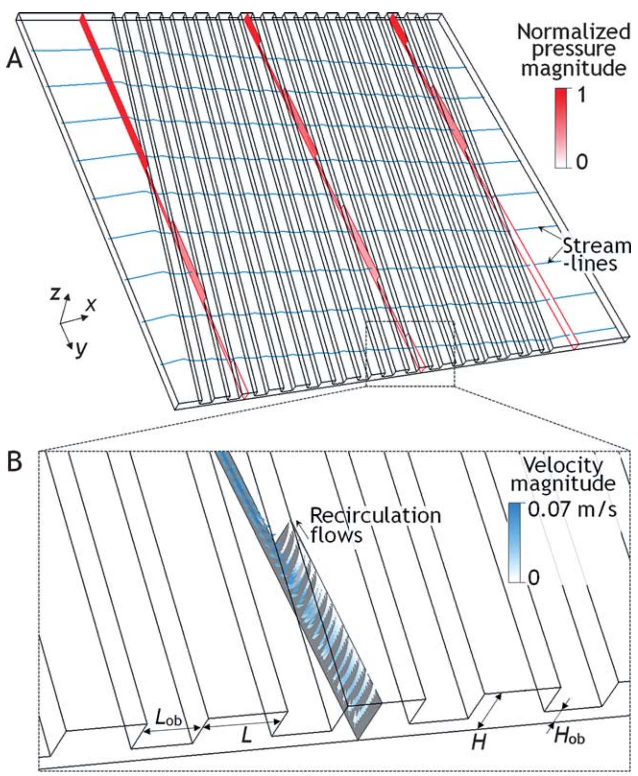

Fig. 2 Numerical simulation of the flow field for 20 μL min^{-1} flow applied to the hydrophoresis devices. Pressure-field and shear-stress magnitudes were calculated for a 3D model of the device with 15 anisotropic obstacles. (A) Simulation of the streamlines (line plot) and pressure fields (colour plot). The device is 1000 μm wide (W) and 22 μm deep (H) with $H_{ob} = 8$ μm, $L_{ob} = 27$ μm, and $L = 33$ μm. (B) Cross-sectional velocity field.

varied from 6.1 to 8.8 × 10^9 cells mL^{-1} and platelets occupied 2.0 to 3.1% of the total cells. The blood sample was diluted 1 : 9 in phosphate buffered saline before platelet separation. Animal care and experimental procedures were performed under the approval of the Animal Care Committees of KAIST.

Experimental setup

The microchannels were imaged with a microscope (TS100; Nikon Co., Japan). A syringe pump (702212; Harvard Apparatus, Holliston, MA) was used to produce 20 μL min^{-1} flows through the single-layer device. Another pump (KDS100; KD Scientific Inc., Holliston, MA) was used to produce 0.2 mL min^{-1} flows through the parallelized device. An image analyzing program, ImageJ (http://rsb.info.nih.gov/ij/), was used to measure particle positions. The program measures the lines drawn by users and converts their pixel information into metric information.

Computational fluid dynamics simulation

The flow characteristics in the wide-channel hydrophoresis device were modeled by using commercial computational fluid dynamics software (CFD-ACE+; ESI Group, Huntsville, AL). The channels for simulation were identical to the grooved channel geometry, but only varied their width from 65 to 1000 μm (Fig. 2). The upwind scheme was used in the conjugates gradient squared (CGS) and preconditioning (Pre) solvers for velocity field, while the algebraic multigrid (AMG) solver was used for pressure correction. The applied flow rate was 20 μL min^{-1} along the flow direction.

Flow cytometry

Flow cytometry (BD LSR II System; BD Biosciences, CA) was used to characterize the sorting performance of the wide-channel hydrophoresis. The forward-light scatter (FSC) and side-light scatter (SSC) profiles are relative measures of cell size and granularity, respectively. Thus, platelets, the smallest, anuclear blood cells were distinguished from RBCs and WBCs on the basis of their distinct FSC and SSC profile.[18] Two gates for cytometric analysis were set around platelets, and WBCs and RBCs, and 10 000 cells per sample were analyzed. The separation purity P was calculated by $P = n_{target}/n_{total} \times 100\%$, where n_{target} is the number of target cells and n_{total} is the total number of cells. The separation recovery R is defined as $R = n_c/n_i \times 100\%$, where n_c is the number of target cells collected in the collection outlet and n_i is the number of target cells injected into the hydrophoresis device.

Results and discussion

Device architecture and physics

Fig. 1 shows the schematics of the wide-channel hydrophoresis device and its size separation principle. We designed the hydrophoresis channel with a low aspect ratio of ≈0.02 in order to lower shear stress level at a given volume throughput of 20 μL min^{-1} and channel height of 22 μm. Thereby, we can decrease maximum wall shear stress up to 20-fold, compared with the hydrophoresis channel with a relatively high aspect ratio of ≈0.34. Since the critical stress level for platelet activation is ≈1000 dyn cm^{-2} at a given exposure time of 10 s,[6] the lowered stress level of 236.3 dyn cm^{-2} and short time period of ≈1 s for separation can ensure platelet purification without significant stimulation. In addition, platelets are suspended in the fluid, not stuck on the surface. The platelets are flowing with the fluid and the velocity gradient at their surface is significantly reduced. As a result, the platelets in the hydrophoresis channel experience the smaller level of shear stress than the wall shear stress, thereby ensuring platelet separation without significant stimulation. We did not observe any platelet aggregation at the contraction–expansion regions. This might be attributed from the smaller difference between the contraction and expansion in dimension and shear rate, compared with the device to monitor platelet aggregation dynamics.[19]

The hydrophoretic microchannel ($H = 22$ μm and $H_{ob} = 8$ μm) is designed so that cells whose diameters range between 4 and 8 μm assume hydrophoretic self-ordering and the microchannel enables to sort the smaller cells, platelet (2 to 4 μm in diameter)[20] in free flow from the large cells (WBCs and RBCs) in hydrophoretic ordering. We have found experimentally that obstacles of $H_{ob} \leq 2D$ (D in particle diameter) hinder the rotational flows of particles and lead to hydrophoretic ordering.[14,15] Even in the wide channel, cell ordering is determined by two kinds of hydrodynamic phenomena: (1) convective vortices induced by the anisotropic fluidics resistance of the anisotropic obstacles, and (2) steric hindrance resulting from the cell–obstacle interaction.[16] Once inside the hydrophoresis device, cells are subject to rotational flows in a counterclockwise direction to the x axis (Fig. 2A and 2B). Such convective vortices make cells move along the transverse direction, following the rotational flows. Steric hindrance occurs when the cells reach near the sidewall 2 in which the cells are subject to the upward flows and begin to collide with the obstacles (Fig. 1B and 2B). This steric effect makes the larger cells (WBCs and RBCs) than the critical diameter diffuse out of the rotational stream and deflects the cells downward, thereby staying near the sidewall 2 (Fig. 1B). In contrast, the smaller cells, platelets can go upward and approach near the channel surface in which the cells are subject to the recirculation flows and begin to deviate from the sidewall 2 (Fig. 2B). In short, the steric effect for platelets is negligible so that they can travel back and forth between two sidewalls without converging, sorting from the larger blood cells.

Device characterization with microparticles and blood cells

The size-separation performance is first characterized with RBCs and polystyrene beads of 4 μm in diameter, to confirm whether the device follows the aforementioned design rule even in the wider channel. Fig. 3A and 3B respectively show the trajectories of RBCs and 4 μm beads obtained at a flow rate of 20 μL min^{-1}. The maximum linear velocity corresponds to ≈0.07 m s^{-1}. Two different patterns of particle flow were observed: RBCs in hydrophoretic ordering and the 4 μm beads in free flow. As going downstream, RBCs begin to migrate across the channel to the equilibrium position, the sidewall 2 (Fig. 3A). At 32 mm downstream, most of RBCs flow within the lateral range from 0 to 200 μm (Fig. 3C). RBCs have a hydrodynamic diameter of

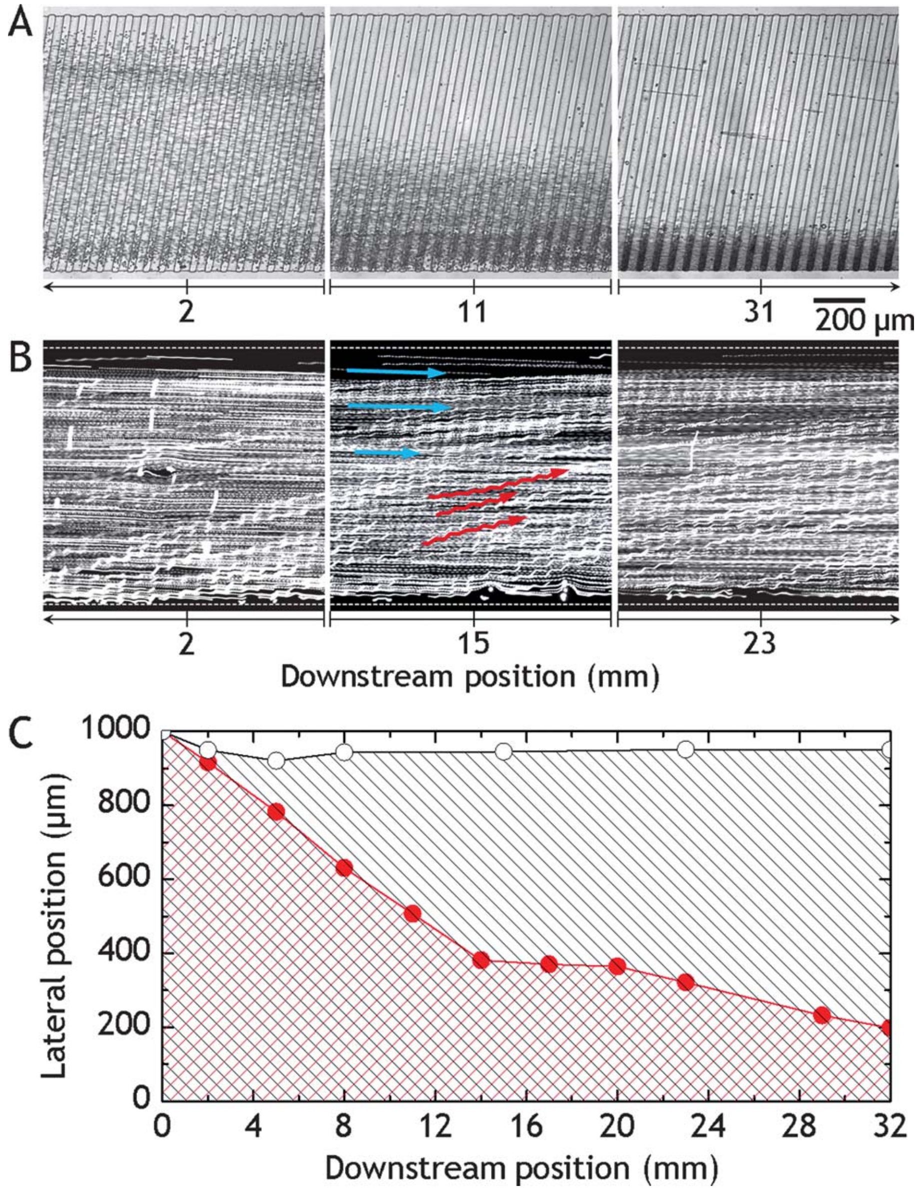

Fig. 3 Size-dependence of hydrophoretic ordering. (A) RBCs initially in a random distribution are focused to the sidewall 2 as going downstream. (B) The 4 μm beads remain unfocused after passing 23 mm of the hydrophoresis channel. The blue and red arrows show the trajectories of the 4 μm beads travelling back and forth between two sidewalls in opposite directions. (C) Ordering patterns in the hydrophoresis device. Each data point represents the outermost position of the major cell- and particle-stream from the sidewall 2: ● for RBCs and ○ for 4 μm beads.

≈7.7 μm,[21] even though they are biconcave disks that have a thickness of 1.7 to 2.6 μm and a disk diameter of 6.2 to 7.9 μm.[22] Presumably, the hydrophoretic channel recognizes the effective diameter over the critical diameter for hydrophoretic ordering rather than the small thickness. In contrast, 4 μm beads have a similar size with the critical diameter so that they are seen to be distributed randomly throughout the channel width, even at 32 mm downstream (Fig. 3B and 3C). These ordering results well support that the hydrophoretic channel enables to sort the small cells, platelet (less than 4 μm in diameter) in free flow from the larger cells (WBCs and RBCs) in hydrophoretic ordering, satisfying the design rule.

The flow conditions in the wide-channel hydrophoresis were tuned from 1 to 20 μL min^{-1}, in order to study the effect of the operating flow speed on hydrophoretic ordering. Flowing particles suspended in fluids are subjected to inertial lift forces such as shear-gradient and wall-induced lift forces. At higher particle Reynolds number (Re_p) of order 1, the inertial lift forces would affect particle behaviors. Re_p is defined as the ratio of the particle inertia to the viscous force; $Re_p = (\rho D^2 U)/(\mu D_h)$, where ρ is the fluid density, D is the particle diameter, U is the maximum fluid velocity, and μ is the dynamic fluid viscosity. D_h is the hydraulic diameter, defined as $D_h = 2WH_{ob}/(W + H_{ob})$. For the highest flow rate of 20 μL min^{-1}, the particle Reynolds numbers are 0.33 and 0.09 for RBCs and platelets, respectively. For low Re_p below 0.33, we did not observe any significant effect of the inertial lift forces on the hydrophoretic ordering of RBCs (see ESI, Fig. S1†).

Channel clogging by blood clots and plaques seriously limits the utility of microfluidic separation devices by disturbing flow distributions and blocking separation channels over time. The use of the wide channel for hydrophoresis prevents channel clogging by blood clots and plaques as well as PDMS debris and dust. Plaques and blood clots move at an angle to the bulk flow and are focused to the sidewall 2 by hydrophoresis like RBCs (see ESI, Fig. S2†). Any channel clogging was not observed for 1 h of device operation. We also did not observe any channel clogging in the device operation even with undiluted blood. This characteristic of the wide channel hydrophoresis enables clogging-free separation of blood cells.

Platelet separation from diluted whole blood

The cell-size dependence of hydrophoretic ordering described above allows continuous, clogging-free sorting of platelets from other blood cell populations (Fig. 4). We quantified the sorting efficiency by collecting blood cells from the lateral range of 500 to 1000 μm of the 1 mm outlet region. The colors in the density plots represent the relative density of cells. The red color represents the highest density of cells and the cells occuring at the lowest frequency is blue in color. After a single round of hydrophoretic sorting, the purity of platelets reached 36.7 ± 3.2% ($n = 3$) from their initial purity of 3.1% at a throughput of 2.0 × 10^5 cells s^{-1} (Fig. 4B and 4C). A second round of hydrophoretic sorting enriched platelets to 82.8 ± 1.2% ($n = 3$) as shown in Fig. 4D. Some of WBCs are too large to pass through the obstacles, while smaller WBCs (D < 8 μm) freely pass through the obstacles and move at an angle to the bulk flow by hydrophoresis. However, the trapped WBCs did not show any significant disturbance to the hydrophoretic sorting (see ESI, Fig. S3†). After each hydrophoretic sorting, platelets were recovered up to ≈41%, compared with their initial concentration. This is attributed to the fact that the half volume of blood was discarded after each separation. The recovery ratio could be further enhanced by adjusting the collection volume ratio between two outlets. The separation purity was significantly affected by the dilution factor (data not shown). Blood volume is normally occupied by RBCs up to 40 to 50%. At the high concentration, RBCs flow adjacent to each other and their cell–cell interaction can affect the separation purity. However, the lowered separation purity could be further improved by integrating serial multiple sorting stages within the microfluidic chip.

The simple geometry and operation of the wide-channel hydrophoresis device facilitate easy parallelization of single separation unit to overcome the limit of throughput performance. Fig. 5A shows a parallelized device comprising 10 stacks of the wide-channel hydrophoresis device. The parallelized

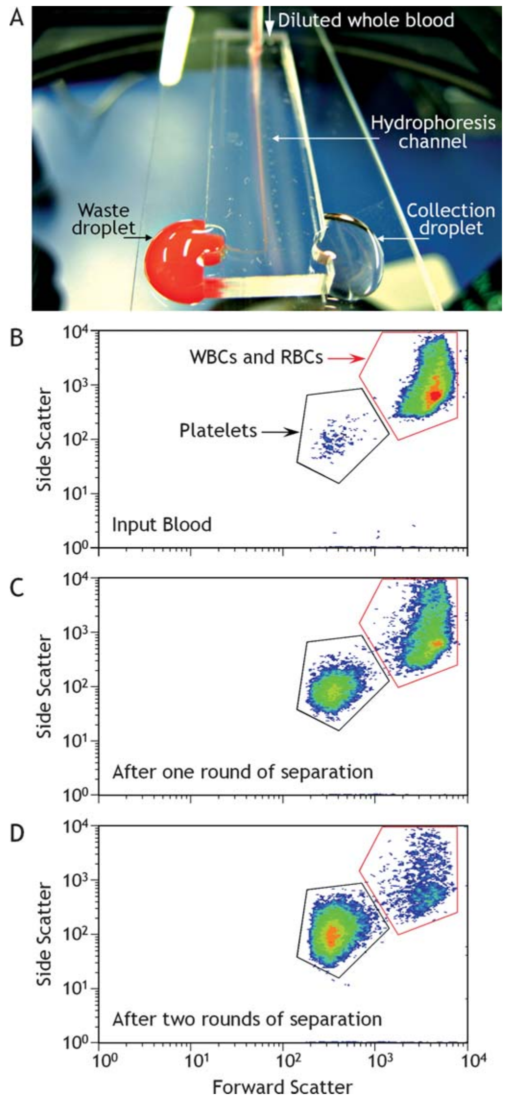

Fig. 4 Hydrophoretic sorting of platelets from diluted whole blood. (A) Photograph showing the operation of the wide-channel hydrophoresis device with a sample inlet, as well as the waste and collection outlet. (B) Unsorted, diluted whole blood sample. (C) Eluted fraction from the collection channel after one round of hydrophoretic sorting. (D) After two rounds of hydrophoretic sorting. Each scatter plot shows a representative result of three independent experiments.

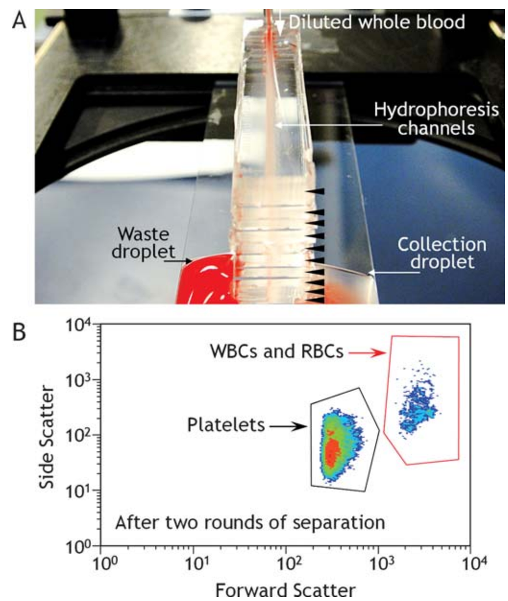

Fig. 5 Parallelization of hydrophoretic sorting. (A) The device consists of 10 channel layers stacked on top of each other. The black triangles denote each channel layer. (B) Eluted fraction from the collection channel after two rounds of hydrophoretic sorting. The fluids fall to the floor and are collected at the two sides.

device augments the throughput capability up to 10-fold and enables high-throughput selection of platelets in physiological shear-stress range. After a single round of hydrophoretic sorting, the purity of platelets reached 29.5% from their initial purity of 2.0% at a throughput of 2.9×10^6 cells s^{-1}. The sorting rate is slightly higher than that of the inertial microfluidic device.[13] A second round of hydrophoretic sorting enriched platelets to 76.8% (Fig. 5B). The throughput capability is only limited by the number of stacked devices and can be further enhanced by stacking more devices, still maintaining physiological shear-stress range.

PRP, an important application of tissue engineering, can be a potential use of this separation method.[23] PRP contains at least seven different growth factors such as platelet-derived growth factors and releases them through degranulation. In order to prepare PRP, platelets should be separated and concentrated normally up to 10 times with extreme care to prevent platelet activation and leukocyte contamination. The current implementation may prove to be useful for platelet separation. However, the additional function to concentrate platelets can be further added simply by integrating a hydrophoresis channel with the lower height of the obstacle gap that platelets assume hydrophoretic ordering.

Conclusions

Here we have successfully designed and characterized a wide-channel hydrophoresis device and its parallelization for high-throughput, clogging-free separation of platelets in physiological shear-stress range. The wide-channel hydrophoresis offers the capability to purify the smallest type of blood cell by the cell-size dependence of hydrophoretic ordering in a low-stress, high-throughput manner. Additionally, this work demonstrates a solution to the challenging problem that current methods in high-throughput cell sorting can be limited by cell stimulation due to the harsh separation condition of high shear stress. The parallelized device comprising 10 stacks of the wide-channel hydrophoresis device augments the throughput capability up to 10-fold, while maintaining physiological shear-stress range. By adjusting the size-separation range of the wide-channel hydrophoresis, it can be further used in separation of rare blood cells enabled by its extreme throughput of 2.9 million cells s^{-1}.

Acknowledgements

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